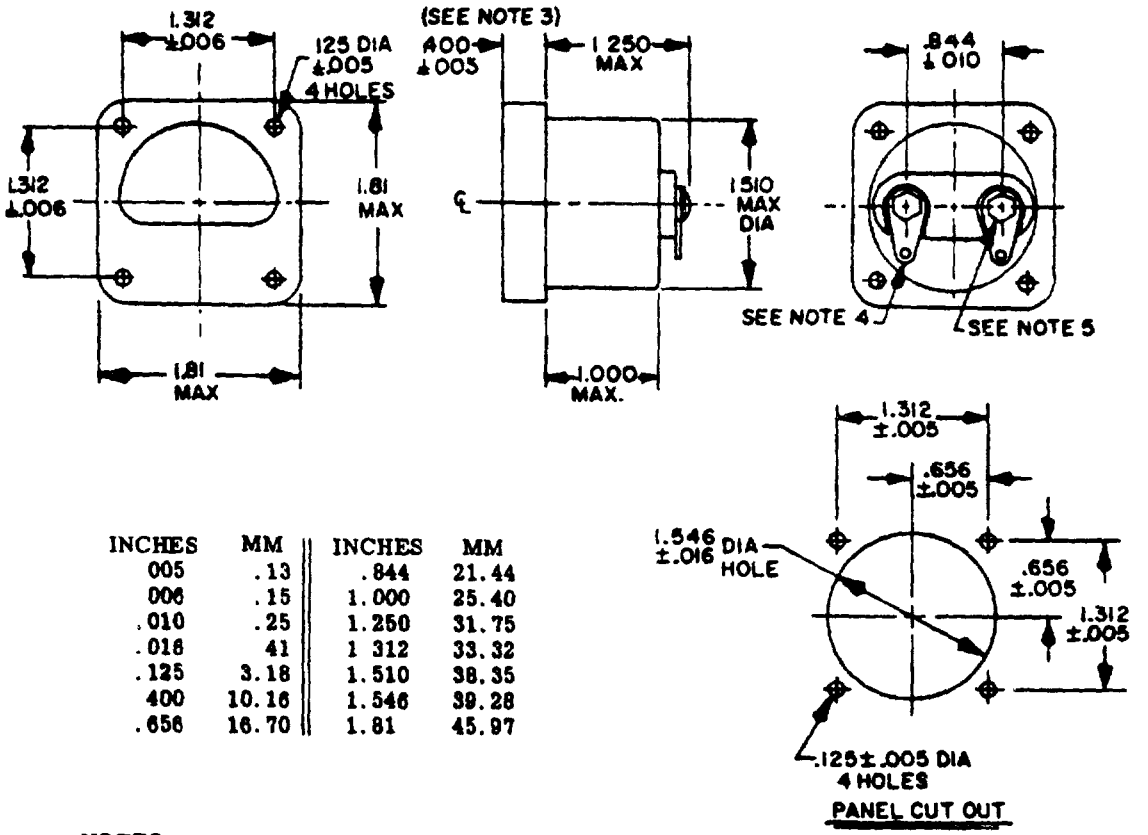
MILITARY SPECIFICATION SHEET

METERS, ELECTRICAL INDICATING, PANEL TYPE, RUGGEDIZED, SPECIAL DECIBEL,

(FLUSH MOUNTING, SQUARE FLANGE, 1-1/2-INCH, STYLE 13)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the meters described herein shall consist of this document and the latest issue of Specification MIL-M-10304.



INCHES	MM	INCHES	MM
005	.13	.844	21.44
006	.15	1.000	25.40
.010	.25	1.250	31.75
.016	41	1 312	33.32
.125	3.18	1.510	38.35
400	10.16	1.546	39.28
.656	16.70	1.81	45.97

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Maximum projection from front of panel, including mounting gaskets (if used).
4. Solder terminals.
5. Equally spaced each side of vertical centerline and within .188 (4.78 mm) inch above or below the horizontal centerline

FIGURE 1. Decibel meter, 1-1/2 inch

Ⓓ denotes changes

FSC 6625

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:50Z
Check the source to verify that this is the current version before use.

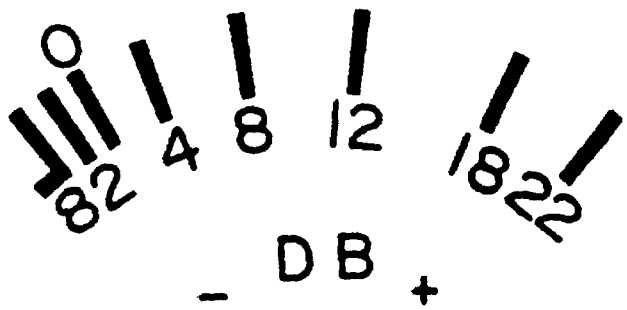


FIGURE 2. Scale plate design

REQUIREMENTS

Dimensions and configuration. See figure 1

Weight: 4 ounces maximum.

Scale length 1 inch minimum.

Color scheme: B.

Pointer Lance type, white non-radioactive.

Range. -8 to +22 dB (see figure 2). Reference level 1 milliwatt in 600 ohms.

Voltage. 0.775 V, 1,000 Hz at 0-dB scale position

Actual level: 0 at 0-dB scale position from zero power level.

Position influence: Not greater than ±2 percent of full scale length for 60 degree rotation from normal operating position.

Initial accuracy. Not to exceed the values shown in table I

(D) Overshoot: 10 percent maximum.

Response time: 0.5 second maximum at 1,000 Hz at the 0-dB scale position.

Effective resistance: 5,000 ohms minimum.

Frequency range: 25 to 10,000 Hz (the change in indication at scale points -8, 0, +8, +12, and +22 dB shall not exceed the values shown in table II).

Special meter Serial number 100.

Heat effect. At 65°C, to be not greater than shown in table III.

Temperature influence. Not greater than shown in table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:50Z
Check the source to verify that this is the current version before use.

Exposure to extreme temperature. Limited to +65°C, change in indication not to exceed the values shown in table V.

Sustained overload. Not to exceed the values shown in table VI

Vibration.
Random drop:
Shock. } Permanent change in indication after each test shall not exceed the val shown in table VII.

Dielectric withstanding voltage 1,500 volts rms, without damage or flashover.

TABLE I Initial accuracies

Scale position	Equivalent voltage with 1,000 Hz sine wave	Error tolerance	Min-max point	Equivalent voltage
(dB)	(Volts)	(dB)	(dB)	(Volts)
-8	0.308	±7	-15 to -1	0 137 to 0 690
-2	0.615	±5	-7 to +3	0.346 to 1.094
0	0.775	±1	-1 to +1	0.690 to 0.869
+4	1.228	±2	+2 to +6	0.975 to 1.542
+8	1.947	±2	+6 to +10	1 542 to 2.449
+12	3.084	±2	+10 to +12	2 449 to 2.882
+18	6.153	±0.75	+17.25 to +18.75	5 653 to 6.700
+22	9.751	±0.50	+21.50 to +22.50	9.210 to 10.320

TABLE II Frequency range

From	To
25 Hz	1,000 Hz ±0.5 dB
1,000 Hz	5,000 Hz ±1 dB
5,000 Hz	10,000 Hz ±3.5 dB

TABLE III Heat effect.

Scale position	Temporary change in indication	Permanent change in indication
-8 dB	±5 dB	±5 dB
0 dB	±2 dB	±1 dB
+8 dB	±1.5 dB	±1 dB
+12 dB	±1.5 dB	±1 dB
+22 dB	±1 dB	±0.5 dB

TABLE IV Temperature influence.

Scale position	Change in indication
-8 dB	±3 dB
0 dB	±2 dB
+8 dB	±2 dB
+12 dB	±2 dB
+22 dB	±1 dB

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:50Z
Check the source to verify that this is the current version before use.

TABLE V. Exposure to extreme temperature.

Scale position	Change in indication (At -55°C)	Permanent change in indication
-8 dB	Free indication	±4 dB
0 dB	Free indication	±2 dB
+8 dB	Free indication	±2 dB
+12 dB	Free indication	±2 dB
+22 dB	Free indication	±1 dB

TABLE VI Sustained overload.

Scale position	Permanent change in indication
-8 dB	±4 dB
0 dB	±1.5 dB
+8 dB	±1 5 dB
+12 dB	±1 5 dB
+22 dB	±1 5 dB

TABLE VII Vibration, random drop, and shock.

Scale position	Permanent change in indication
-8 dB	±6 dB
0 dB	±3 5 dB
+8 dB	±2 dB
+12 dB	±2 dB
+22 dB	±2 dB

Custodians:
Army - EL
Navy - EC
Air Force - 82

Review activities
Army - EL, AV, AT, ME, WC
Navy - EC, MC
Air Force - 11
DSA - ES

User activities:
Army - MI
Navy - CG
Air Force - None

Preparing activity
Army - EL

Agent
DSA - ES

(Project 6625-0574-14)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:50Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street City State ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 82 MAR

PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:50Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements All users of military standardization documents are invited to provide suggestions This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems Enter in block 6 any remarks not related to a specific paragraph of the document If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered

NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12062 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US ARmy Avionics Research and Development Activity
Attn DAVAA-S/PE
Fort Monmouth NJ 07703

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:50Z
Check the source to verify that this is the current version before use.